Introduction to Programming

Lecture 9

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Programming Toolkit

- Decisions
- Loops
- Sequences

Laboratory Stool

Constructing a laboratory Stool

Constructing a laboratory Stool

- Task: Making a stool
  - Subtask:
    - Make a seat
    - Make legs for the stool
    - Assemble them

What we will study today ...

- What are functions?
- How are they defined ?
- How are they declared ?
- What values are passed to functions ?
- What values do functions return ?

Function

Function name

{

Body of the function

}

Function

Two types of functions:

- Functions that return a value
- Functions that do not return a value

Function

```
return-value-type  function-name( argument-list )
{
   declarations and statements
}
```

Declaration of Function

```
return-value-type  function-name( argument--type-list) ;

main ( )
{
  :
}
```

Example

```
int function-name ( int , int , double ) ;

void main ( )
{
    ....
}
```

Definition of Function

```
int function-name ( int i , double j )
{
    ...
}
```

Return Type of Function

Declaration

```
int square ( int ) ;
```

Definition

```
int square ( int i )
{
  return ( i * i ) ;
}
```

Function Call

```
int x ;
x = square ( i ) ;
```

Example: Function to calculate integer power (X^n)

```
double raiseToPow ( double x , int power )
 {
  double result ;
  int i ;
  result = 1.0 ;
  for ( i = 1 ; i <= power ; i ++ ) // braces first
  {
   result * = x ;  // result  = result  *x
  }
  return ( result ) ;
 }
```

Code to Call the raisetopow Function

```
include < iostream.h >
void main ( )
{
 double x ;
 int i ;
 cout << " Please enter the number " ;
 cin >> x ;
 cout << " Please enter the integer power that you want this number raised to " ;
 cin >> i ;
 cout << x << " raise to power  " << i << "is equal to " << raiseToPow ( x , i ) ;
}
```


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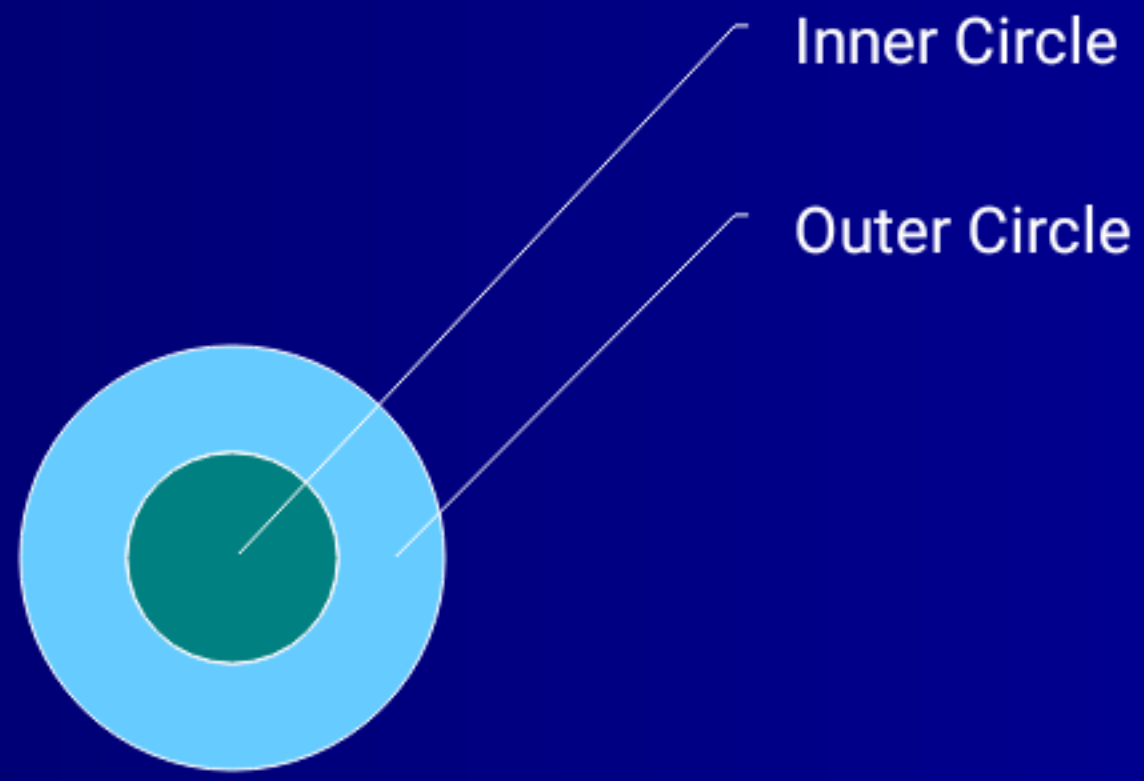
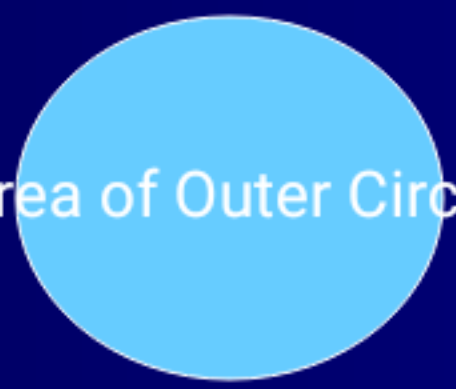
Call By Value

Calling function

Called function

Area of the Ring





Area of Outer Circle —— Area of Inner Circle = Area of the Ring

Example: Function to calculate the area of a circle

```
double circleArea ( double radius )
{
    return ( 3.1415926 * radius * radius ) ;
}
```

Calculating ringArea without using Function

```
main ( )
{
    :
    :
  ringArea = ( 3.1415926 * rad1 * rad1 ) – ( 3.1415926 * rad2 * rad2 ) ;

}
```

Exercises

- Modify the ***raise to power*** function so that it can handle negative power of x, zero and positive power of x.

- For the ***area of ring*** function put in error checking mechanism.

In today's lecture

- We used functions for breaking complex problems into smaller pieces, which is a top-down structured approach.
- Each function should be a small module, self contained and it should solve a well defined problem.
- Variable names and function names should be self explanatory.
- Always comment your code

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